

DO NOT WRITE ON TEST



**INFORMATION
TECHNOLOGY CONCEPTS**

~OPEN EVENT~

(391)

NATIONAL 2022

TOTAL POINTS _____ **100 Points**

Test Time: 60 minutes

MULTIPLE CHOICE

Identify the choice that *best* completes the statement or answers the question.

1. When using database terminology, _____.
 - a. a table is the same as an entity set
 - b. a record is the same as an attribute
 - c. a field is the same as an entity
 - d. none of these are correct
2. The relational field _____.
 - a. does not have to have the same name in both tables
 - b. must have the same name in both tables
 - c. must be unique in both tables
 - d. is not necessary to relate two tables in a DBMS
3. Kerberos uses which of the following?
 - a. Symmetric key cryptography
 - b. Asymmetric key cryptography
 - c. Federated identity
 - d. Multi-factor authentication
4. In a network, which protocol controls the right to send a message?
 - a. UDP
 - b. CSMA/CD
 - c. TCP
 - d. FTP
5. Which of the following is *not* an Internet application?
 - a. FTP
 - b. Email
 - c. Telnet
 - d. CERT
6. The “point” at which one network is linked to an internet is often called a(n) _____ because it serves as a passageway between the network and the outside world.
 - a. opening
 - b. gateway
 - c. portal
 - d. bridge
7. In the TCP/IP hierarchy, which layer chops messages into Internet-compatible units?
 - a. Application
 - b. Transport
 - c. Network
 - d. Link

8. _____ is the premeditated, politically motivated attack against information computer systems, computer programs, and data that results in violence against noncombatant targets by subnational groups or clandestine agents.
- Cyberterrorism
 - Ecoterrorism
 - Cyber Espionage
 - None of these
9. Data can be stolen or destroyed during a(n) _____ attack.
- money center
 - memory
 - economic
 - hardware
10. The joint task force representing the four components of the U.S armed services is the _____.
- U.S. Cyber Command
 - Patriot Command
 - Cyber Terror Association
 - None of these
11. Which of the following incidents fits into the severity level 2 category?
- Malware detected on multiple systems
 - User's excessive use of bandwidth or resources
 - Compromise or suspected compromise of any company website or web presence
 - Compromise or suspected compromise of protected customer information
12. In what of the following situations would a denial-of-service attack occur?
- Clicking a link in an e-mail that happens to contain malware
 - Installing a keylogger on a system in an attempt to collect usernames and passwords
 - Flooding a system with ping requests in an attempt to take it down
 - Viewing medical records to satisfy one's curiosity
13. Once a class is defined, _____.
- it may be referenced as a single variable
 - a sub class may be defined
 - an object may be defined
 - all of these are correct
14. To sum the rows and columns of a two-dimensional array, the _____.
- rows are summed, then the columns, and finally the grand total
 - columns are summed, then the rows, and finally the grand total
 - rows and columns are summed together
 - rows and columns are summed in the same inner loop with the results placed in an added row and an added column. The grand total is summed in the outer loop.

15. An administrator can limit how much space a user can use in each file system by enabling a _____.
a. quota check
b. mount option
c. snapshot
d. disk quota
16. Which of the following is a way to test software design?
a. Entity-relationship diagram
b. Class diagram
c. Structure chart
d. Structured walkthrough
17. NFS requires another service called RPC, which stands for which of the following?
a. Reverse Procedure Call
b. Remote Portmap Call
c. Remote Procedure Call
d. Reverse Portmap Call
18. Which of the following is a network storage solution based on the communication protocol provided by SCSI?
a. WWID
b. LUN
c. iSCSI
d. Initiator
19. Using the _____ command, you can discover SAMBA servers in your workgroup.
a. testparm
b. smbclient
c. smbpasswd
d. nmblookup
20. Which of the following best describes the goal of the network layer of the OSI model?
a. It is responsible for addressing the packets.
b. It defines the characteristics of the network hardware.
c. It is responsible for establishing connections and terminations between the computers.
d. It is responsible for the transfer of data.
21. Which of the following keeps track of the state of the connection?
a. Packet filters firewalls
b. Routers
c. Stateful inspection firewalls
d. All these are correct

22. Which of the following statements about packet filter firewalls is not true?
- a. They can filter traffic based on the port number.
 - b. They can filter traffic based on the protocol type.
 - c. They can filter traffic based on the state of the connection.
 - d. They can filter traffic based on the source and destination address.
23. Translation of data from a virtual machine to a physical server is facilitated by the _____.
a. hardware
b. memory
c. hypervisor
d. network interface card
24. In TCP/IP, which layer determines which direction a message segment is sent on the Internet?
a. Application
b. Transport
c. Network
d. Link
25. The _____ model is also a popular means of distributing files such as music recordings and motion pictures via the Internet.
a. Peer-to-peer
b. Open source
c. Business-to-business
d. Client/server
26. A packet-switched network IP is used by which generation of wireless networks?
a. 3G
b. 4G
c. 5G
d. All of these are correct
27. _____ is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources
a. IaaS
b. Service aggregation
c. Resource pooling
d. Cloud computing

28. A cloud_____ is an entity that manages the use, performance, and delivery of cloud services, and negotiates relationships between CSPs and cloud consumers.
- a. carrier
 - b auditor
 - c. broker
 - d. provider
29. _____ storage uses a flat address space instead of the hierarchical, directory-based scheme.
- a. Block
 - b. File-based
 - c. Object
 - d. Network attached
30. An algorithm that falls into which of the following categories is generally considered more efficient?
- a. $(\log_2 n)$
 - b. (n)
 - c. $(n \log_2 n)$
 - d. (n^2)
31. Which of these is not an algorithmic representation?
- a. Stepwise refinement
 - b. Pseudocode
 - c. Flowchart
 - d. Programming language
32. What attribute is common to all access controls that identifies unique records in a set, such as a username?
- a. Identification scheme
 - b. Authentication method
 - c. Authorization model
 - d. None of these choices
33. Which of the following is not an example of knowledge-based authentication?
- a. Cognitive password
 - b. PIN
 - c. One-time passcode
 - d. Password
34. What is an example of providing your mother's maiden name as verification?
- a. Multilayer authentication
 - b. One-time passcode
 - c. Multi Factor authentication
 - d. Cognitive password

35. Which of the following commands is used to view the permissions of a file or directory?
- a. ls -l
 - b. cd
 - c. chmod
 - d. umask
36. Which of the following commands is used to specify which default permissions to exclude when creating a new file or directory?
- a. chmod
 - b. cd
 - c. umask
 - d. ls
37. SUID refers to which of the following special permission sets?
- a. Specify User ID
 - b. Set Umask ID
 - c. Set User ID
 - d. Specify Umask ID
38. Which of the following is a key activity performed during the testing phase of the SDLC?
- a. Map security and privacy requirements
 - b. Threat modeling and design review
 - c. Final security review and application security monitoring
 - d. Security test cases and dynamic analysis
39. Which if the following best describes an account takeover attack?
- a. Websites that sell credit cards numbers, PINS, Social Security numbers and other personality identifiable information
 - b. Using stolen credit card data to transact
 - c. Using stolen log-in credentials to access private bank or credit card account
 - d. An email that appears to be from a legitimate bank asking the user to click a link to confirm their login credentials
40. What is an example of a technical control that can be used for physical security?
- a. Audit trails/access logs
 - b. Fire suppressions systems
 - c. HVAC
 - d. Mantraps
41. Buffers are _____.
- a. data to be transferred from one storage device to another storage device
 - b. only used when processing files sequentially
 - c. data holding areas in the computer
 - d. all of these are correct

42. The fields of a record contain _____.
a. like data for all records
b. different pieces and number of data items for each record
c. only character or string data
d. only numeric data
43. When using linked lists, _____.
a. new records are always added to the bottom of the file
b. deleting data is accomplished by moving all records up, after the deleted record, therefore, eliminating the deleted record
c. when data is deleted, the links are changed to jump over the deleted record. The deleted record number is entered into the empty list
d. new records are always added to the top of the file
44. SELinux refers to which of the following?
a. Security Enabled Linux
b. Security Enhanced Linux
c. System Enhanced Linux
d. System Enabled Linux
45. Which of the following operations security controls help mitigate the effects of a loss?
a. Preventive controls
b. Detective controls
c. Corrective controls
d. Deterrent controls
46. How does a program sanitize a disk by writing 1s and 0s?
a. Formatting
b. Deleting
c. Degaussing
d. Overwriting
47. Access control that is discretionary is best described by which of the following?
a. The information owner decides who has access to resources.
b. Access to resources is determined based on the need to know principle.
c. Access to resources is decided by the system based on the concept of subjects, objects, and labels.
d. Users are grouped based on a common access need, such as job function.
48. One or more applications can be accessed through one centralized interface of which type?
a. Remote access portal
b. Virtual private network
c. DMZ
d. Network access control system

49. In what of the following scenarios is multifactor authentication used?
- a. A password and a PIN
 - b. A cognitive password and a PIN
 - c. A fingerprint scan and a retina scan
 - d. A smartcard and a retina scan
50. Which of the following components of an operating system is not part of the kernel?
- a. User Interface
 - b. File manager
 - c. Scheduler
 - d. Boot Record